RAJALAKSHMI ENGINEERING COLLEGE

Shwetha Ram R 251801318-2025-AIDS S2

Started on	Thursday, 6 August 2026, 11:08 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 11:50 AM
Time taken	19 days
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      int count =0;
5      int rev=0;
6      int remainder;
7      count++;
8      count ++;
9
10     scanf("%d",&n);
11     while(n !=0){
12         count++;
13         remainder =n%10;
14         count++;
15         rev = rev*10 +remainder;
16         count++;
17         n/=10;
18         count++;
19     }
20     count++;
21     printf("%d",count);
22     return 0;
23 }
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Shwetha Ram R 251801318-2025-AIDS S2

Started on	Thursday, 6 August 2026, 10:59 AM
State	Finished
Completed on	Tuesday, 25 August 2026, 2:12 PM
Time taken	19 days 3 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main(){
    int n;
    int count =0;
    int c=0;
    count++;
    scanf("%d",&n);
    count++;
    for(int i=n/2;i<n;i++){
        count++;
        for(int j=1;j<n;j=2*j){
            count++;
            for(int k=1;k<n;k=k*2){
                c++;
                count++;

                count++;
            }
            count++;
        }
        count++;
    }
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Shwetha Ram R 251801318-2025-AIDS S2

Started on	Thursday, 6 August 2026, 10:50 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:59 AM
Time taken	9 mins 34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include<stdio.h>
int main(){
    int num;
    int count =0;
    scanf("%d",&num);
    count ++;
    for(int i=1;i<= num;++i){
        count++;
        if(num%i==0){
            count++;

        }
        count++;
    }
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Shwetha Ram R 251801318-2025-AIDS S2

Started on	Thursday, 6 August 2026, 10:31 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:49 AM
Time taken	18 mins 45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int main(){
    int n;
    int count = 0;
    scanf("%d",&n);
    count ++;
    if(n==1){
        count++;

    }
    else{
        for(int i=1;i<=n;i++){
            count++;
            for(int j =1;j<=n;j++){
                count ++;

                count ++;
                count ++;
                break;
            }
            count ++;
        }
        count ++;
    }
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

Shwetha Ram R 251801318-2025-AIDS S2

Started on	Thursday, 6 August 2026, 10:17 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:45 AM
Time taken	28 mins 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n;
    int i=1;
    int s =1;
    int count=2;
    scanf("%d",&n);
    while(s<=n){
        count ++;
        i++;
        count ++;
        s += i;
        count ++;
    }
    count ++;
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course